CEFET/RJ

DAL & Comunidade

PACOTES E AJUDA

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Pacotes

- Pacotes são coleções de funções disponibilizados como bibliotecas
  - Publicados no repositório CRAN
    - Controle de qualidade
    - Boa documentação
  - Carregados de um repositório GitHub
    - Versões em desenvolvimento
- Atualmente têm mais de 18500 pacotes
  - https://cran.r-project.org/
- R tem uma comunidade muito ativa
  - Vários pesquisadores, professores, programadores e estatísticos
- Três pacotes são do Data Analytics Lab
  - https://cran.r-project.org/web/packages/TSPred/index.html
  - https://cran.rstudio.com/web/packages/gstsm/index.html
  - https://cran.r-project.org/web/packages/STMotif/index.html

Instalação de pacotes

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R*

Source on Save | Run | Source

```
#funções gráficas qplot
?qplot

library(ggplot2)

?qplot

install.packages("ggplot2")
```

8:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from C:/R/main/.RData]

> ?qplot
No documentation for 'qplot' in specified packages and libraries:
you could try '??qplot'
> library(ggplot2)
> ?qplot
>
```

Environment History Connections Tutorial

Import | 132 MiB | List

R | Global Environment

Environment is empty

Files Plots Packages Help Viewer Presentation

R: Quick plot | Find in Topic

qplot {ggplot2} R Documentation

Quick plot

Description

Deixe comentado o install.packages para não ficar rodando várias vezes a instalação

Uso de função do pacote ggplot2

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R

Source on Save | Run | Source

```r
x <- c(-1, -0.8, -0.6, -0.4, -0.2, 0, 0.2, 0.4, 0.6, 0.8, 1)
x

y <- x^3
y

qplot(x, y)
```

1:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> x <- c(-1, -0.8, -0.6, -0.4, -0.2, 0, 0.2, 0.4, 0.6, 0.8, 1)
> x
 [1] -1.0 -0.8 -0.6 -0.4 -0.2  0.0  0.2  0.4  0.6  0.8  1.0
>
> y <- x^3
> y
 [1] -1.000 -0.512 -0.216 -0.064 -0.008  0.000  0.008  0.064
 [9]  0.216  0.512  1.000
>
> qplot(x, y)
>
```

Environment History Connections Tutorial

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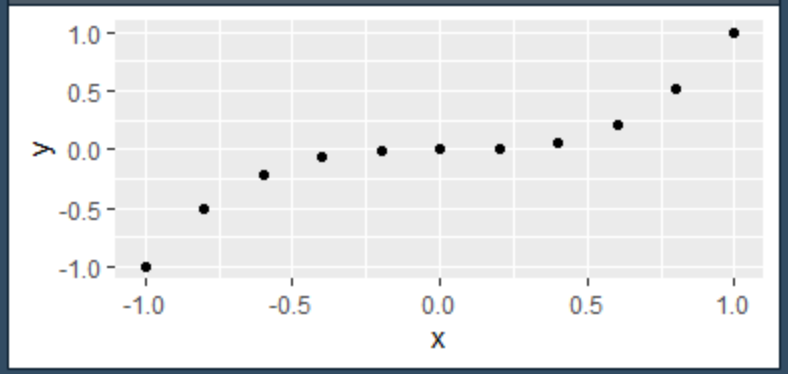
values

x	num [1:11] -1 -0.8 -0.6 -0.4 ...
y	num [1:11] -1 -0.512 -0.216 -...

Files Plots Packages Help Viewer Presentation

Zoom | Export



Chamadas com parâmetros explícitos não precisa seguir a ordem os argumentos

Prototipando uma função

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R

Source on Save | Run | Source

```
1
2  # criar uma função para jogada
3  dado <- 1:6
4  dados <- sample(dado, size = 2, replace = TRUE)
5  sum(dados)
6
7
8
9
```

8:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> source("C:/Users/eduar/OneDrive/Aulas/R/02-Pacotes-Ajuda.R")
>
```

Environment History Connections Tutorial

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R Global Environment

values

dado	int [1:6] 1 2 3 4 5 6
dados	int [1:2] 4 5

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modified
..		
.RData	122 B	Sep 19,
.Rhistory	26.5 KB	Sep 27,

Prototipando uma função

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R*

Source on Save | Run | Source

```
1  jogada <-  function() {
2    dado <- 1:6
3    dados <- sample(dado, size = 2, replace = TRUE)
4    return(sum(dados))
5  }
6
7  jogada()
```

1:1 | jogada() | R Script

Console | Terminal | Background Jobs

R 4.2.1 · C:/R/main/

```
> source("C:/Users/eduar/OneDrive/Aulas/R/02-Pacotes-Ajuda.R")
> jogada <-  function() {
+   dado <- 1:6
+   dados <- sample(dado, size = 2, replace = TRUE)
+   return(sum(dados))
+ }
>
> jogada()
[1] 7
>
```

Environment | History | Connections | Tutorial

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Functions

jogada	function ()

Files | Plots | Packages | Help | Viewer | Presentation

C: > R > main

Name	Size	Modified
..		
.RData	122 B	Sep 19,
.Rhistory	26.5 KB	Sep 27,

Criando a função

Clique na vassoura para limpar as variáveis

Simulação "mínima"

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R*

Source on Save | Run | Source

```r
simulacao_minima <- replicate(30, jogada())
qplot(simulacao_minima, binwidth =1)
mean(simulacao_minima)
```

1:1 (Top Level) R Script

Environment History Connections Tutorial

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Values	
simulacao_...	int [1:30] 10 7 6 10 6 8 3 6 ...
Functions	
jogada	function ()

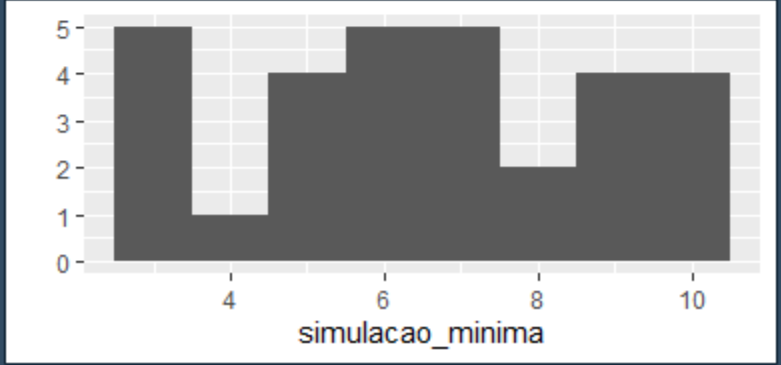
Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> simulacao_minima <- replicate(30, jogada())
> qplot(simulacao_minima, binwidth =1)
> mean(simulacao_minima)
[1] 6.533333
>
```

Files Plots Packages Help Viewer Presentation

Zoom | Export



O número mínimo de amostras para calcular uma média é 30, mas, pode não ser suficiente para apresentar uma distribuição ...

Espaço de possibilidades

						(6,1)					
combinations					(5,1)	(5,2)	(6,2)				
of die 1 and				(4,1)	(4,2)	(4,3)	(5,3)	(6,3)			
die 2 that add			(3,1)	(3,2)	(3,3)	(3,4)	(4,4)	(5,4)	(6,4)		
up to sum		(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(3,5)	(4,5)	(5,5)	(6,5)	
	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)	(2,6)	(3,6)	(4,6)	(5,6)	(6,6)
	2	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
						sum					

Veja o cenário das possibilidades (probabilidades)
https://rstudio-education.github.io/hopr/packages.html

Simulação "completa"

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R*

Source on Save | Run | Source

```
simulacao_completa <- replicate(10000, jogada())
qplot(simulacao_completa, binwidth =1)
mean(simulacao_completa)
```

1:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> simulacao_completa <- replicate(10000, jogada())
> qplot(simulacao_completa, binwidth =1)
> mean(simulacao_completa)
[1] 7.0006
>
```

Environment History Connections Tutorial

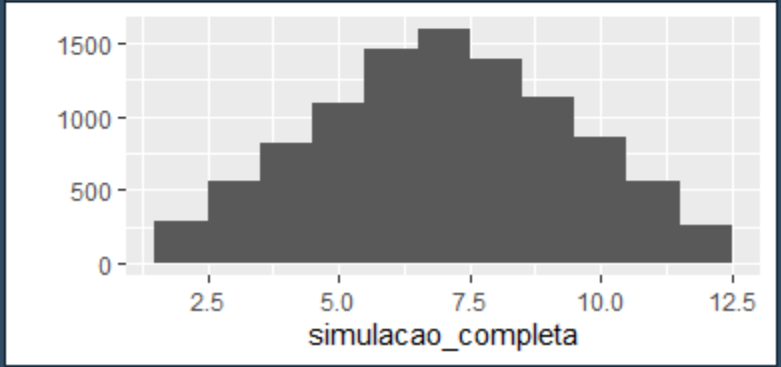
Import | 186 MiB | List

R | Global Environment

Values	
simulacao_...	int [1:10000] 6 5 4 7 8 10 4 ...
simulacao_...	int [1:30] 10 7 6 10 6 8 3 6 ...
Functions	
jogada	function ()

Files Plots Packages Help Viewer Presentation

Zoom | Export



Ajuda para raiz quadrada

R: Miscellaneous Mathematical Functions Find in Topic

MathFun {base} R Documentation

Miscellaneous Mathematical Functions

Description

`abs(x)` computes the absolute value of `x`, `sqrt(x)` computes the (principal) square root of `x`, $\sqrt{x}$.

The naming follows the standard for computer languages such as C or Fortran.

Usage

```
abs(x)
sqrt(x)
```

Arguments

`x` a numeric or `complex` vector or array.

Details

These are internal generic primitive functions: methods can be defined for them individually or via the `Math` group generic. For complex arguments (and the default method), `z, abs(z) == Mod(z)` and `sqrt(z) == z^0.5`.

`abs(x)` returns an `integer` vector when `x` is `integer` or `logical`.

S4 methods

Both are S4 generic and members of the `Math` group generic.

References

Becker, R. A., Chambers, J. M. and Wilks, A. R. (1988) *The New S Language*. Wadsworth & Brooks/Cole.

See Also

`Arithmetic` for simple, `log` for logarithmic, `sin` for trigonometric, and `Special` for special mathematical functions.

'plotmath' for the use of `sqrt` in plot annotation.

Examples

Run examples

```
require(stats) # for spline
require(graphics)
xx <- -9:9
plot(xx, sqrt(abs(xx)),  col = "red")
lines(spline(xx, sqrt(abs(xx)), n=101), col =
```

[Package *base* version 4.2.1 Index]

Usando o código da ajuda

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R

Source on Save | Run | Source

```
1 require(stats) # for spline
2 require(graphics)
3 xx <- -9:9
4 plot(xx, sqrt(abs(xx)),  col = "red")
5 lines(spline(xx, sqrt(abs(xx)), n=101), col =  "pink")
```

1:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> require(stats) # for spline
> require(graphics)
> xx <- -9:9
> plot(xx, sqrt(abs(xx)),  col = "red")
> lines(spline(xx, sqrt(abs(xx)), n=101), col =  "pink")
>
```

Environment History Connections Tutorial

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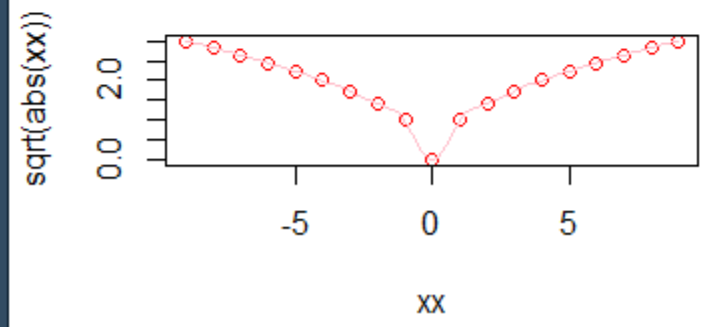
R | Global Environment

Values

xx	int [1:19] -9 -8 -7 -6 -5 -4 ...

Files Plots Packages Help Viewer Presentation

Zoom | Export



Explore os exemplos da documentação

Ajuda para amostra

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R*

Source on Save | Run | Source

```
1 ?sample
2
```

2:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> require(stats) # for spline
> require(graphics)
> xx <- -9:9
> plot(xx, sqrt(abs(xx)),  col = "red")
> lines(spline(xx, sqrt(abs(xx)), n=101), col =  "pink")
> ?sample
>
```

Environment History Connections Tutorial

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R | Global Environment

Values

xx	int [1:19] -9 -8 -7 -6 -5 -4 …

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R: Random Samples and Permutations | Find in Topic

x	either a vector of one or more elements from which to choose, or a positive integer. See 'Details.'
n	a positive number, the number of items to choose from. See 'Details.'
size	a non-negative integer giving the number of items to choose.
replace	should sampling be with replacement?
prob	a vector of probability weights for obtaining the elements of the vector being sampled.
useHash	logical indicating if the hash-version of the

Versão da amostragem com distribuição de probabilidades

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R*

Source on Save | Run | Source

```r
simulacao = replicate(10000,
                      sample(2:12,
                      replace=TRUE,
                      prob=c(1,2,3,4,5,6,5,4,3,2,1)/36))
qplot(simulacao, binwidth=1)
mean(simulacao)
```

6:16 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> simulacao = replicate(10000,
+                       sample(2:12,
+                       replace=TRUE,
+                       prob=c(1,2,3,4,5,6,5,4,3,2,1)/36))
> qplot(simulacao, binwidth=1)
> mean(simulacao)
[1] 7.000945
>
```

Environment History Connections Tutorial

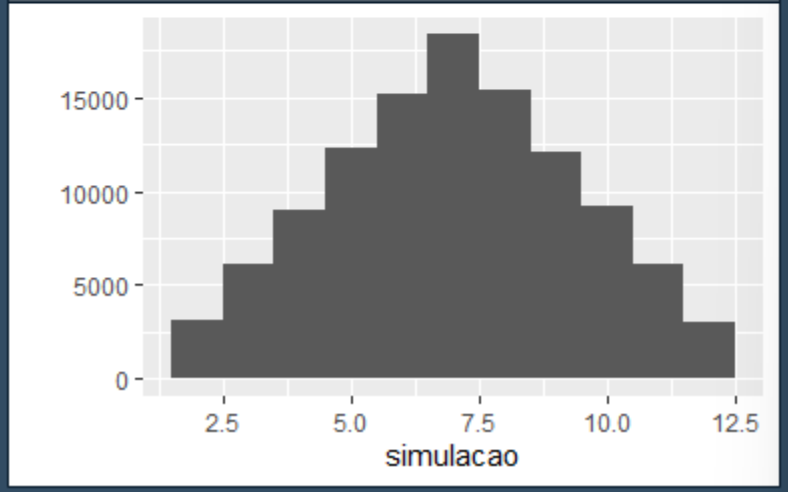
Import | 334 MiB | List

R | Global Environment

Data	
simulacao	int [1:11, 1:10000] 7 8 9 5...
Values	
xx	int [1:19] -9 -8 -7 -6 -5 -4 ...

Files Plots Packages Help Viewer Presentation

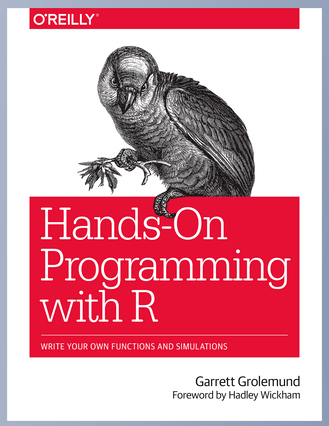
Zoom | Export

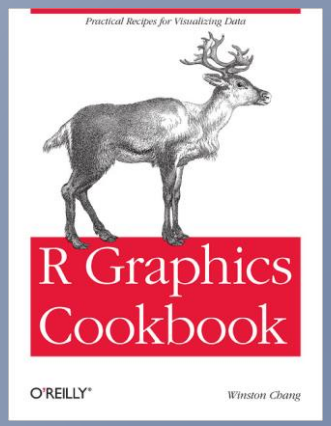


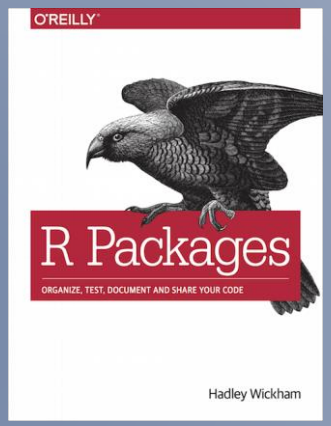
Usando a distribuição de probabilidade para cada valor possível

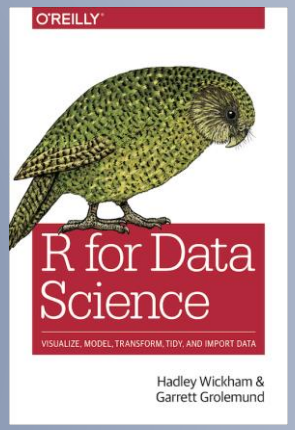
Referências

Código: https://nbviewer.org/github/eogasawara/R/blob/main/02-Pacotes-Ajuda.ipynb









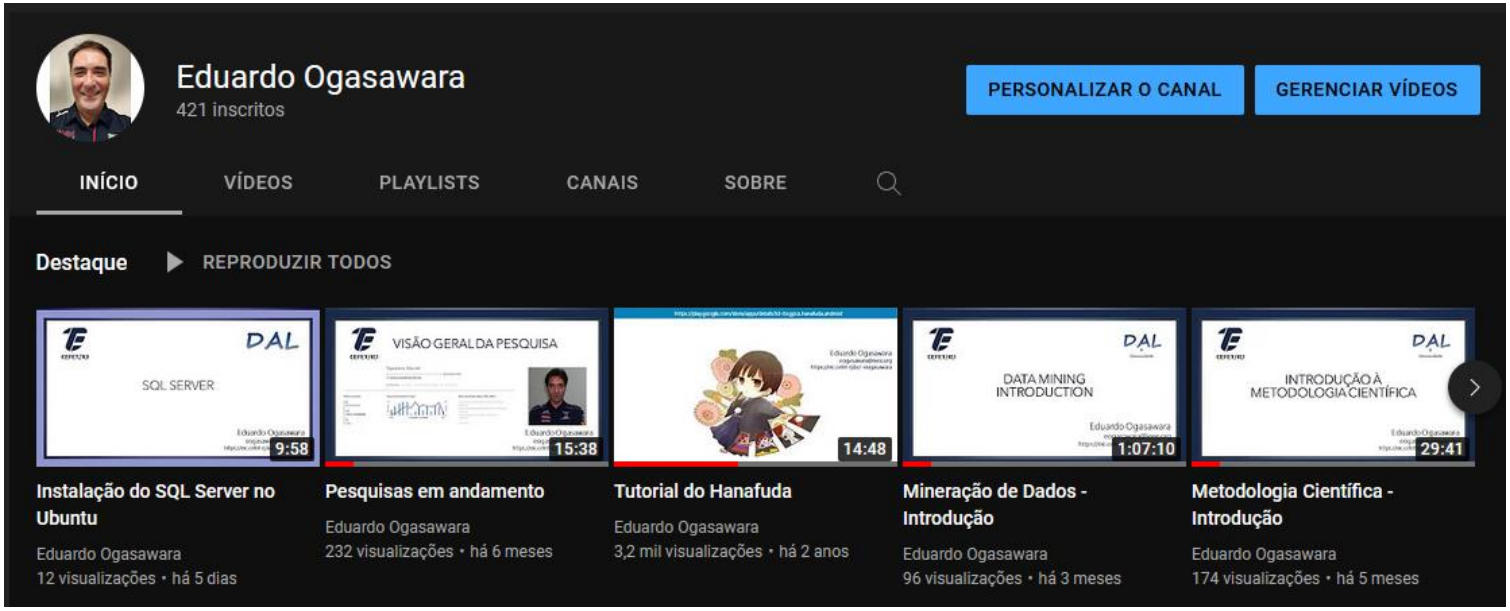
Hands-on Programming with R: https://rstudio-education.github.io/hopr/index.html
R Graphics Cookbook: https://r-graphics.org
R Packages: https://r-pkgs.org/index.html
R for Data Science: https://r4ds.had.co.nz

Leitura adicional: https://rstudio-education.github.io/hopr/packages.html

[1] G. Grolemund, 2014, Hands-On Programming with R: Write Your Own Functions and Simulations. O'Reilly Media, Inc.

Mantenha-se atualizado

Inscreva-se:



Slides and Jupyter notebooks: